Answer **one** question from this section.

6 **(a)** Define *electric field strength* at a point.

……

……

…… [1]

(b) **(i)** A long positively charged rod is placed at a distance from a stationary -2.0 μC point charge as shown in Fig. 6.1.

Sketch the electric field pattern between the rod and the point charge. [1]

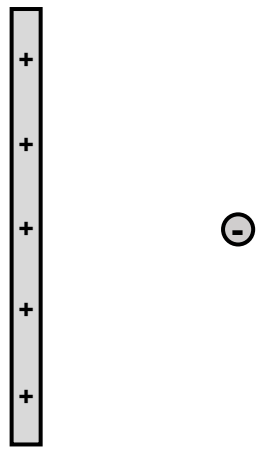

Fig. 6.1

(ii) A current I is passed through the rod and the charge moves at a speed of 1.5 cm s^{-1} as shown in Fig. 6.2. The charge experiences a magnetic field strength of 10 mT.

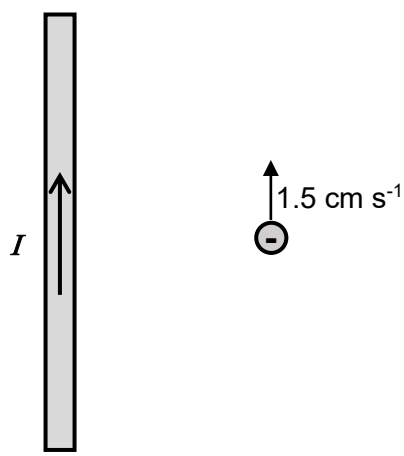

Fig. 6.2

For Examiner's Use

1. Calculate the magnitude of the force experienced by the charge due to the magnetic field of the rod.

force = ……. N [2]

2. The negative charge moves in a curved path to the right.

Explain this observation.

……

……

……

……

…… [2]

3. State and explain a modification that can be made to Fig. 6.2 such that the negative charge can be made to move parallel to the rod when the current I is still passing though the rod.

……

……

……

……

…… [2]

For Examiner's Use

(iii) In a second experiment, the rod carrying a current of 4.0 mA is placed next to a horizontal flat coil of wire which is fixed in its position. The centre of the coil, P, is at a perpendicular distance of 20 cm from the rod as shown in Fig. 6.3.

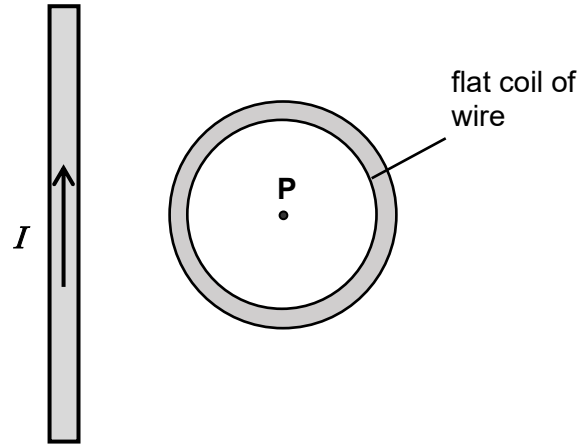

Fig. 6.3

1. Calculate the magnitude of the magnetic flux density at P.

magnetic flux density = ……. T [2]

2. A very small current now flows in the flat coil of wire in an anti-clockwise direction.

Without any further calculation, state and explain if the magnetic flux density at P will change compared to that obtained in **(b)(iii)1**.

……

……

……

……

…… [2]

For Examiner's Use

3. State and explain the direction of the net force acting on the coil.

……

……

……

……

…… [2]

(c) A circular coil is placed with its axis vertical and a bar magnet aligned with the axis of the coil, is held above the coil as shown in Fig. 6.4. The bar magnet is then dropped. A datalogger is connected to the coil and records the e.m.f. induced in the coil.

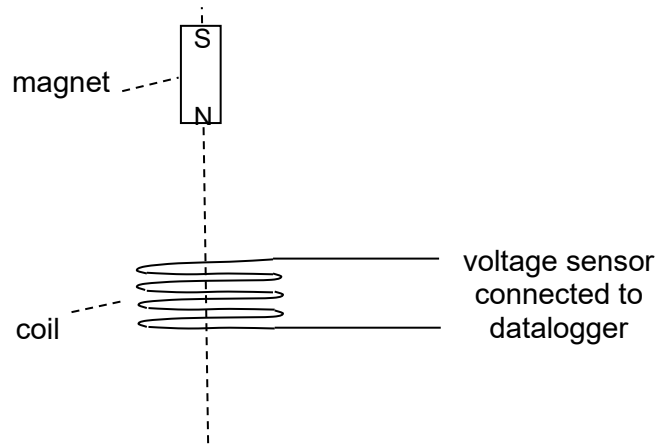

Fig. 6.4

Fig. 6.5 shows the variation with time of the e.m.f. induced as the magnet falls through the coil.

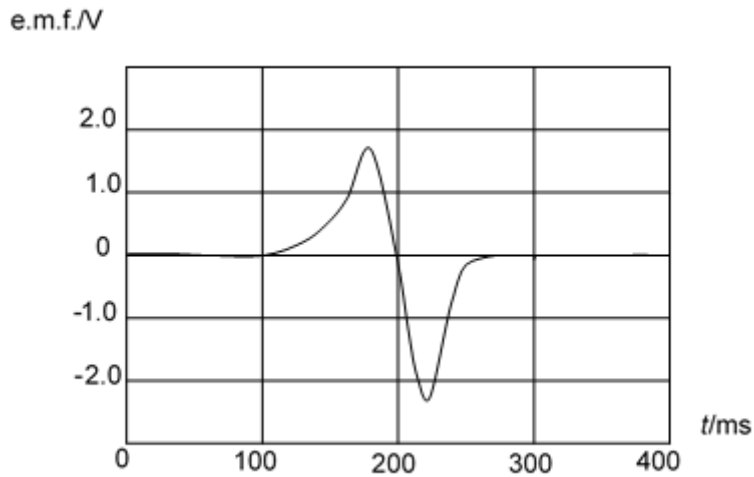

Fig. 6.5

For Examiner's Use

(i) State what the area under the graph between 200 ms and 270 ms represent.

……

……

…… [1]

(ii) Using the laws of electromagnetic induction, explain the shape of the graph obtained in Fig. 6.5.

……

……

……

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……

……

…… [3]

(iii) The experiment is repeated with the magnet now dropped from a higher height.

State and explain how the graph in Fig. 6.5 will change.

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……

…… [2]